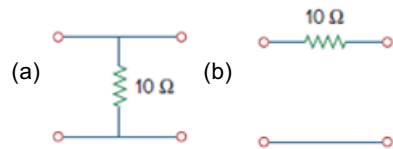


Problem

If the two single-element two-port networks in Fig. 19.64 are cascaded, then **D** is:

- (a) 0
- (b) 0.1
- (c) 2
- (d) 10
- (e) undefined

Figure 19.64



Step-by-step solution

Step 1 of 5

Consider the following circuit diagrams:

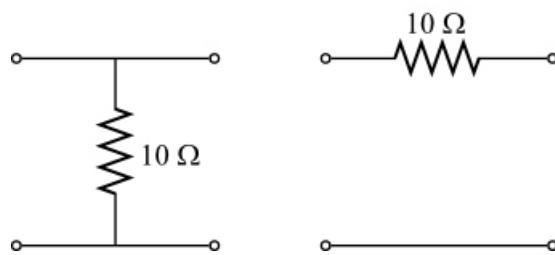


Figure 1

Step 2 of 5

The two single element circuits in the Figure 1 are cascaded draw the modified circuit diagram:

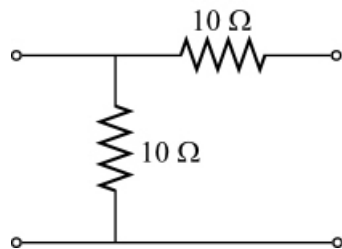


Figure 2

Step 3 of 5

To calculate the **D** parameter, connect a DC test source to input port and short circuit the output port. Draw the modified circuit diagram:

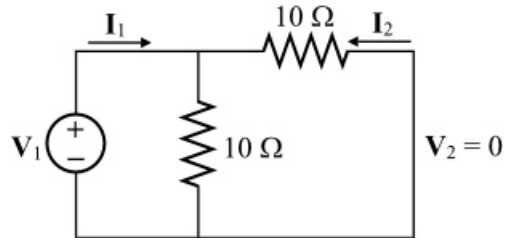


Figure 3

Step 4 of 5

Calculate the value of current I_1 .

$$\begin{aligned} I_1 &= \frac{V_1}{10} + \frac{V_1}{10} \\ &= \frac{2V_1}{10} \\ &= \frac{V_1}{5} \end{aligned}$$

Calculate the value of current I_2 .

$$I_2 = -\frac{V_1}{10}$$

Step 5 of 5

Calculate the **D** parameter.

$$\begin{aligned} D &= -\frac{I_1}{I_2} \\ &= -\frac{\left(\frac{V_1}{5}\right)}{\left(-\frac{V_1}{10}\right)} \\ &= -\left(\frac{V_1}{5}\right)\left(-\frac{10}{V_1}\right) \\ &= 2 \end{aligned}$$

Therefore, the value of **D** parameter is 2.

Therefore, the correct option is ☒ c.